Lesson 2: How Much Energy Does It Take to Move?

Unit

Energy: Stores and Transfers

Lesson focus

Students calculate the energy in the kinetic store with $E_k = ½ m v^2$, convert grams to kilograms, and explain why doubling the speed gives four times the energy.

Specification

AQA GCSE Physics 4.1.1.2 Changes in energy. The equation is on the AQA Physics equation sheet; students still learn what each symbol means and how to use it.

Scope

Students calculate E_k from mass and speed. They are **not** expected to find m or v from a given E_k in this unit (finding v needs a square root).

Learning objectives

By the end of the lesson, students should be able to:

1. State that the kinetic store depends on mass and speed.
2. Calculate kinetic energy using $E_k = ½ m v^2$, with the correct units.
3. Convert a mass in grams to kilograms.
4. Explain why doubling the speed gives four times the kinetic energy.

Key vocabulary

Keyword	Student-friendly definition
Kinetic store	The energy an object has because it is moving
Mass (m)	How much matter an object has, in kilograms (kg)
Speed (v)	How fast an object moves, in metres per second (m/s)
Squared	A number multiplied by itself: $3^2 = 3 \times 3 = 9$

The equation

Quantity	Symbol	Unit
Kinetic energy	E_k	joule (J)
Mass	m	kilogram (kg)
Speed	v	metre per second (m/s)

$$E_k = ½ m v^2$$

kinetic energy = 0.5 × mass × speed²

Only the speed is squared. Check this on every calculator: students who type 0.5 × m × v and then square the whole answer get the wrong result.

Equipment and resources

- A trolley or toy car, a ball, a 1 m marked track, stopwatches
- A top-pan balance (in grams)
- Calculators
- Optional: light gate and data logger

Starter: Which would hurt more?

Show pairs and ask which has more energy in its kinetic store, and why:

1. A tennis ball at 20 m/s, or a bowling ball at 20 m/s?
2. A car at 10 m/s, or the same car at 30 m/s?
3. A lorry at 2 m/s, or a bullet at 400 m/s?

Students usually get 1 and 2. The third is the hook for the lesson: the answer is not obvious until you calculate.

Main teaching: The equation, step by step

Always write the equation, then substitute, then calculate.

Worked example 1:

> *A 1200 kg car moves at 15 m/s. Calculate the energy in its kinetic store.*
> $E_k = ½ m v^2$
> $E_k = 0.5 \times 1200 \times 15^2$
> $E_k = 0.5 \times 1200 \times 225$
> $E_k = 135\,000$ J

Worked example 2 (converting grams):

> *A 450 g football is kicked at 20 m/s. Calculate the energy in its kinetic store.*
> $m = 450$ g $= 0.450$ kg
> $E_k = ½ m v^2$
> $E_k = 0.5 \times 0.450 \times 20^2$
> $E_k = 90$ J

Main teaching: Double the speed, four times the energy

Repeat worked example 1 at 30 m/s:

> $E_k = 0.5 \times 1200 \times 30^2 = 540\,000$ J

540 000 ÷ 135 000 = 4. Ask students to explain why: the speed is squared, so $2^2 = 4$. Then predict, before calculating, what tripling the speed does (9 times).

Real-world link: this is why a crash at 60 mph is far worse than at 30 mph, and why stopping distances grow so quickly with speed. The brakes must transfer four times as much energy to the thermal store.

Doubling the **mass** only doubles the energy, because mass is not squared.

Activity 1: Calculation page

Students complete the workbook calculation page: straight substitution first, then grams to kilograms, then comparison questions ("how many times bigger?"). Include the lorry and bullet from the starter:

Object	Mass	Speed	E_k
Lorry	30 000 kg	2 m/s	60 000 J
Bullet	10 g	400 m/s	800 J

Activity 2: Measure it (practical)

Groups find the energy in the kinetic store of a trolley rolling along the bench.

1. Measure the trolley's mass on the balance in grams; convert to kilograms.
2. Mark a 1.00 m section of track.
3. Push the trolley gently so it rolls through the section; time it with a stopwatch.
4. Speed = distance ÷ time.
5. Calculate E_k. Repeat three times and find the mean speed.

Sample results: mass 500 g, times 1.62 s, 1.55 s, 1.59 s; mean 1.59 s; speed 1.00 ÷ 1.59 = 0.63 m/s; $E_k = 0.5 \times 0.500 \times 0.63^2 \approx 0.10$ J.

Discussion: the answer is tiny. Compare it with the car. Which measurement has the biggest effect on the result, and why? (The time: an error in speed is squared.)

Check for understanding

1. Which two quantities does the kinetic store depend on?
2. What unit must mass be in?
3. Calculate the energy in the kinetic store of a 60 kg runner at 5 m/s. (750 J)
4. A car's speed halves. What happens to the energy in its kinetic store? (A quarter of the energy remains.)

Plenary

Exit ticket:

1. E_k = _____
2. 250 g = ____ kg
3. If the speed doubles, the kinetic energy is multiplied by ____, because ____.

Possible prep

Find the masses and typical speeds of three moving things (an animal, a vehicle, a ball in a sport) and calculate the energy in each kinetic store. Rank them.

Teacher notes

Common mistakes:

1. Squaring everything, or squaring nothing. Make students write v^2 as a separate line the first few times.
2. Using grams. Ask "is the mass in kilograms?" before every substitution.
3. Thinking twice as fast means twice the energy.

Speed = distance ÷ time is assumed from Key Stage 3. If students are shaky, give them the speeds and skip the practical's timing.